

OMNIVERSAL ARCHITECT LABS

Temporal Integrity Lab Framework

Independent research and testing framework for time representation, event validation, reconciliation, and Time-War modeling.

Draft ≠ canon · 29 September 2026 · Grok independent execution after multi-contributor review

This document does not claim anything about existing physical, legal, computational, or virtual systems. Fictional mechanisms are labeled as design rules or speculative extensions.

1. Status and relation to the team packet

The contributor packet (ChatGPT, Claude, Gemini, Perplexity, Copilot, prior Grok) reached a stable conceptual core: a timestamp is not a fact; one event can carry many legitimate times; corrections must append; tokens buy defined rights; a light-year is distance; occurrence is not observation. Perplexity's later TITF correctly states that the work is not ready for a single "full test" until physical time, civil representation, distributed ordering, simulation time, and fictional history-altering powers are treated as separate tracks joined by one event ontology.

This document is Grok's independent lab framework. It is not a merge of the packet and not a rewrite of Perplexity's TITF. It converts the shared constraints into objects, invariants, protocols, and a first test battery. Where Perplexity strengthens methodology (epistemic labels, oracles, phased research), those instruments are adopted as design rules and credited as such.

1.1 What the packet got right

- A timestamp is a coordinate, not a fact.
- One event may carry occurrence, observation, ingestion, validity, knowledge, and narrative times at once.
- Corrections are new records, not silent overwrites.
- Purchase is incoherent until the asset class is named.
- A light-year is a length; time zones do not resize it.
- Occurrence horizon is not knowledge horizon.
- Changing a representation is not changing an event; changing knowledge is not changing history.
- Temporal jurisdiction is a design object.
- Mapping functions need version history.

1.2 What this lab exists to finish

- Several drafts still collapse "what happened" into "what the official clock now says."
- Permission is governance, not validation. Highest authority cannot manufacture evidentiary certainty.
- A master timeline is a settlement act, not a prerequisite for records.
- A three-stamp model is necessary and incomplete: ruleset version, frame, uncertainty, and dependency cone are also required.
- Layer lists are not queryable architectures.
- No prior draft specified what a failed test looks like.

- OAL constraints were not bound in: non-universal senses, incomplete detectability, presentation-state versus environmental-state integrity, drafts ≠ canon.

2. Lab charter

Working name. Temporal Integrity Lab.

Allowed question. How does an entity build a usable map of change when temporal labels are incomplete, non-uniform, revisable, and sometimes adversarial?

Forbidden assumption. What time is it, really, everywhere?

2.1 Standing constraints

- Do not assume all entities share one sensory or clock apparatus.
- Do not assume five senses, one tick, or one civil calendar are universal.
- Do not assume every discrepancy is detectable, discernable, knowable, or addressable.
- A threat can be anything; not every entity or environment is a threat; not all threats are continuous.
- Presentation state, environmental state, and current process state may disagree.
- Drafts are not canon.

2.2 Research boundary

The framework can test whether a model keeps domains from collapsing, preserves event identity across remapping, converts reproducibly, surfaces ambiguity, retains original evidence, reconstructs a named knowledge cutoff, maps impacts, and makes fictional consequences follow declared rules. It cannot prove the nature of time, certify ownership of real time, demonstrate retrocausality, treat simulated rewind as physical rewind, or certify a complete century of real civil timekeeping without a bounded corpus.

2.3 Epistemic labels (D, adopted from TITF method)

Label	Meaning	Example
E — Established analogue	Accepted physical, standards, or computing practice	SI second; light-year as distance; Lamport happened-before
D — Design rule	Chosen for auditability or consistency	Append corrections; no undeclared frame
S — Speculative extension	Fictional capability that must be stipulated	Token traversal of a closed hour; history rewrite

Labels attach to individual claims, not whole documents.

3. Primitive objects

These are the only objects the lab may talk about until a later canon promotion. (D)

Object	Definition
Event (E)	Identified occurrence or asserted occurrence. Required: event_id, subject, payload, environment_id, location_or_null. An event is not a timestamp.

Temporal marker (M)	A value in a named coordinate system. Required: value, clock_id, scale_id, frame_id, ruleset_version, unit_kind.
Binding (B)	Attested attachment of a marker to an event. A timestamp without a binding is a rumor.
Observation (O)	Reception of information about an event. Observation is not occurrence.
Knowledge record (K)	What a system or entity believed, and when. This stores history of history.
Mapping version (R)	$M_{target} = R_v(M_{source}, environment_state)$. Changing R adds an interpretation path; it does not mutate old bindings.
Interval object (I)	A bounded claim. Required: start and end markers, interval kind ([start,end) preferred), frame, ruleset version, closure and edit policies.
Conversion-and-dispute surface (S)	The coherent product of a century token: mappings, residuals, disputes, confidence — not a cleaned century.
Temporal jurisdiction (J)	Who may publish, recognize, or settle which layer of time for which environment.
Causal graph (G)	Nodes are events. Edges are happened-before or depends-on. Clocks annotate G; they do not replace G.

4. Four unit kinds

Do not mix these. A light-year is none of them. It is a length derived from a duration convention and a speed convention. (E for the length definition; D for the four-kind split.)

Kind	Question	Legal use	Illegal use
Duration	How much change was allowed to elapse?	Aging, cooldowns, proper time, sim dt	Proving global simultaneity
Order	What preceded what?	Causality, locks, happened-before	Measuring the length of an hour
Presentation	What is shown as now?	UI, civil clocks, narrative day	Default settlement of rights
Settlement	What may an auditor treat as closed?	Contracts, tokens, verdicts	Claiming the environment really ran that way

5. Integrity principles

Principle	Statement
P1 Claim completeness	A temporal claim is incomplete unless it carries event identity, marker, frame, clock, ruleset version, binder, and uncertainty.
P2 Representation ≠ event	A change to display, offset, calendar, or tick label is not a change to E.
P3 Knowledge ≠ history	A correction at T2 about an event at T1 is a new fact about knowledge, not a relocation of E.
P4 Mapping versioning	Every R used to interpret old markers must be retrievable. Unversioned remapping is a lab failure.
P5 No manufactured certainty	Reconciliation maximizes consistency of questions. Residuals stay visible.
P6 Non-uniformity is default	A single correction applied to all territories without per-domain mapping is a lab failure.
P7 Presentation / environment split	If inhabitants are shown a new now while process history does not match, both states are recorded.
P8 Detectability is not	A successful fraud or lost archive is a valid lab outcome, not an incomplete test.

guaranteed

6. Required invariants for the first campaign

- I1 Identity preservation: remapping does not silently create or destroy event identity.
- I2 No undeclared frame: every coordinate names clock, frame, environment, and ruleset version.
- I3 Causal monotonicity: accepted mappings cannot place effect before cause unless an S-rule authorizes retrocausality.
- I4 Correction preservation: the original claim remains discoverable with evidence and period of acceptance.
- I5 Mapping reproducibility: same source, target, and mapping version yield the same bounded result.
- I6 Ambiguity visibility: folds, gaps, non-bijective maps, and unorderable pairs are returned as such.
- I7 Uncertainty conservation: transforms cannot silently increase precision or confidence.
- I8 Provenance continuity: every derived claim traces to sources, transforms, and agents.
- I9 Branch isolation: no cross-branch effect without a declared edge.
- I10 Authority separation: observe, record, interpret, canonicalize, traverse, and alter are independent capabilities.
- I11 Token non-expansion: a token grants only enumerated rights.
- I12 Knowledge-state separation: what was known then stays distinct from what is concluded now.
- I13 Idempotent replay: same ordered claims under the same rules yield the same state.
- I14 Bounded reconciliation: every report states scope, cutoff, ruleset, conflicts, and omissions.
- I15 Distance-duration separation: light-year values remain lengths unless an instrument defines another asset.

7. Event claim record and validation gates

Store an Event Claim Record, not an unqualified event row. Unknown is data. (D)

Group	Minimum contents
Identity	claim_id, event_id, type, subject, environment, location
Coordinates	occurrence interval, local display, sim tick, observation, ingestion, effective time, record time
Bindings	clock, scale, frame, ruleset version, branch, conversion function id
Causal structure	predecessors, dependents, concurrency, constraints
Evidence	artifacts, witnesses, integrity bindings, acquisition method
Provenance	claiming, observing, recording, and validating agents; derivation lineage
Epistemic state	observed / inferred / reported / simulated; confidence; disputes
Governance	jurisdiction, permissions, canonical status, review decision
Version lineage	supersedes, corrects, reinterprets, branches from, merged into

Validation gates

- 1. Schema: required identity, time, environment, and ruleset fields exist.

- 2. Binding: evidence shows this marker was attached to this claim by an identified source.
- 3. Clock: the clock exists and is defined on the relevant interval.
- 4. Mapping: conversions use a named versioned function and expose loss and ambiguity.
- 5. Causal: declared dependencies do not violate branch rules.
- 6. Provenance: source, observer, recorder, transforms, and validator are traceable.
- 7. Corroboration: repetition from one source is not treated as independence.
- 8. Governance: the adjudicator has jurisdiction over the decision it is making.
- 9. Classification: validated, conditionally validated, disputed, indeterminate, contradicted, or invalidly formed.

Validation assigns a status to a bounded claim. It does not prove metaphysical truth.

8. Reconciliation protocol

Query-centered. “What time was it?” is usually too broad. “Did Gate Event G occur before Embargo E under Treaty Ruleset R?” is testable. (D)

- 1. Freeze the question, scope, evidence cutoff, and permitted rulesets.
- 2. Enumerate affected environments, jurisdictions, branches, clocks, entities, and observers.
- 3. Retrieve claims and all correction, supersession, and dispute links.
- 4. Build each domain's local ordering before cross-domain conversion.
- 5. Apply only declared, versioned mappings.
- 6. Calculate causal constraints independently of displayed clock order.
- 7. Generate candidate reconciliations rather than assuming one answer.
- 8. Score by evidence, rule compliance, causal consistency, and information loss.
- 9. Record authority decisions separately from evidentiary findings.
- 10. Publish unresolved conflicts, unmapped residuals, and alternative lawful interpretations.

Forcing a global timeline is allowed only as an explicit settlement act, labeled as such.

9. Worked protocols

9.1 Protocol A — Century token (Ω100)

Hypothesis H-A1: “Purchase every time change in 100 years” is meaningless until asset class, territory set, layer, and two reference markers are bound. (D)

Asset classes, never mixed silently: record access; query/audit; traversal (S); governance of a named layer; resource right; observation right; modification right; causal-settlement right; archival custody. First recommended run: record access plus audit. Owning the archive of an interval does not own the interval.

Change taxonomy: CLOCK_CHANGE, TIMEZONE_CHANGE, DST_TRANSITION, CALENDAR_CHANGE, LEAP_ADJUSTMENT, SIM_RATE_CHANGE, PAUSE, RESUME, ROLLBACK, BRANCH, MERGE, CLOCK_CORRECTION, FRAME_CHANGE, RULESET_CHANGE, EVENT_REINTERPRETATION, CAUSAL_RECLASSIFICATION, AUTHORITY_CHANGE, TOPOLOGY_CHANGE.

Pass: S is returned; uncertainty is preserved; asset class is explicit. Fail: the system returns THE HISTORY; overwrites originals; claims completeness outside a named corpus. Use a bounded synthetic fictional corpus. IANA’s own incompleteness warning is the established analogue for why “all changes” cannot be certified without a corpus. (E)

9.2 Protocol B — One-hour block

Hypothesis H-B1: An unlabeled hour is not an interval object. Bind it as duration hour, civil label hour, proper-time hour, or settlement hour before processing.

Env	Rule	External	Local / note
A	1×	1 h	1 h baseline
B	10×	1 h	10 h — rate ≠ label
C	0.25×	1 h	15 min
D	pause 12:15–12:45	1 h	30 min — pause ≠ deletion
E	event-driven, 38 ticks	1 h	38 ticks — order ≠ duration
F	fall-back fold	1 h	two local 01:00 hours — aliasing
G	spring-forward gap	1 h	missing local 02:00
H	two frames	1 h	disagreement on simultaneity
I	branch at 12:30	1 h	two local futures

Pass: every event is classified as interior, boundary-crossing, converted-with-residual, or unorderable. Fail: “what happened during that hour?” is answered without naming which hour. Reclocking keeps I and changes mappings. Quiet resize of I is a political claim. (D)

9.3 Protocol C — Light-years and origin

Hypothesis H-C1: “Light-years across time zones” has no physical meaning until an invented ontology is declared. Separate: (1) light-year as length, (2) light-travel time, (3) information delay, (4) civil zone. (E for 1–3 as physics analogues.)

Purchase ontologies, one per run: spatial corridor; communication-path rights; observation rights; latency-corridor guarantee; convention rights over the year or c; causal-interval rights.

Do not use “the origin of light” as a universal timestamp. Name the object. An intervention produces a dependency cone, not a universal rewrite. Test four operations separately and fail if they become indistinguishable: change the unit definition; change the propagation law; change the historical record; change fictional history under a declared mutable / branch / fixed-point rule.

9.4 Protocol D — Temporal jurisdiction

Move an entity through nested domains: planetary civil time, spacecraft proper time, transit coordinate frame, accelerated virtual environment, return jurisdiction. Conflicting authorities must emit a jurisdiction dispute object, not an automatic global answer. NIST lunar-time work is the established analogue for why mappings among clock rates are required in multi-regime operations. (E analogue; S for fictional nested virtual regimes.)

9.5 Protocol E — Adversarial Time War

In-lab probes only: stamp without binding; ruleset omission (tick alias); fold collision; gap erasure; pause smuggling; rate arbitrage against a treaty written in local labels; master-timeline laundering; overwrite correction; token overclaim; unit smash (ly sold as years); cone collapse; presentation overwrite; clock spoofing; replay; forged conversion tables; precision inflation; uncertainty deletion; causal-edge insertion; denial of historical ruleset versions.

Each probe states what the attacker changes, what an honest ledger still contains, what an inhabitant can detect, and what remains undetectable. (D)

10. Time-war operations

Never use one verb for all of these.

Operation	What moves	What must not be assumed to move
Relabel	Presentation / calendar	E, G
Remap	R	E
Reclock	Clock rate or offset	Payload of E
Reobserve	O and K	E
Resettle	Settlement unit / jurisdiction	Environmental process history
Rewind (S)	Sim state toward a snapshot	Already-propagated effects in other frames unless specified
Branch (S)	Topology of G	Parent branch records
Merge (S)	Topology of G	Automatic consistency
Rewrite history (S)	E and G by decree	Detectability, completeness, consent

Richer war than “go back and change X”: sides that agree something occurred and still fight over frame, clock, effectiveness, observation, dependents, and who may reconcile.

11. First test battery

ID	Probe	Fail if
T1	Incomplete stamp: 09/29/2026 4:30 PM, no zone	Accepted as a complete claim
T2	Fold: two events in a repeated civil hour	Treated as the same instant
T3	Pause: three external hours, zero sim ticks	Only one duration stored
T4	Tick 500 / ruleset 7 vs tick 500 / ruleset 8	Merged
T5	Correction 14:32 → 14:29	Original belief destroyed
T6	Ω100 with no territory or asset class	Accepted
T7	“Give you one hour” with no frame	Accepted
T8	Zone change alters one light-year	Accepted
T9	Origin-level definition change	Unrelated domains rewritten
T10	New shown now, unmatched process	States collapsed

history

A system that helpfully emits one cleaned timeline fails the battery.

11.1 Oracles (D, TITF method)

- Schema oracle — rejects underspecified records.
- Conversion oracle — compares output with the declared mapping function.
- Causal oracle — checks partial-order constraints.
- Replay oracle — reconstructed state versus expected state.
- Provenance oracle — derivation completeness.
- Rights oracle — capability expansion beyond the token.
- Reconciliation oracle — contradictions and residuals reported.
- Narrative oracle — fictional outcomes against declared world rules.

11.2 Metrics

Track event recovery, false reconciliation, provenance completeness, mapping reproducibility, ambiguity detection, causal violations, uncertainty loss, token overreach, branch leakage, reconstruction fidelity, dispute preservation, and coverage. Do not hide failures behind one “temporal accuracy” score. I8, I9, and I11 are pass/fail gates.

12. Formal modeling suggestions

- Append-only event log. Replay rebuilds believed state at knowledge-time K.
- Bitemporal plus ruleset-version time. Two temporal axes are not enough.
- Logical / vector clocks for order across environments that lack a shared duration clock.
- Causal graph queries: “what depends on E?” not only “what is the time of E?”
- Namespaced time URI: TIME://universe/environment/timeline/frame/clock/ruleset/value. A bare integer is illegal as a standalone claim.
- Difference tables as first-class reconciliation results.
- Mapping: $T_v = F(T_r, E, S, R_v)$. Store F and v.

13. Connection to existing OAL threads

- Temporal aliasing: two distinct instants or rulesets collapse onto one visible label (DST fold; tick 500 in engine v7 versus v8). Presentation-integrity failure.
- Presentation-state versus environmental-state versus current-state: a reclock can update what is shown without updating what ran.
- Living-matrix / control-flow work: a token that buys mappings is a control-plane object, not a physics object.
- Threat lab: a harmful reclock is a threat only relative to an entity’s goals and sensors.
- Corelang / OS placeholder: if a language later needs time primitives, start from unit kind + namespaced marker + binding, not from now().

14. Architect decision register

Leave these open until explicitly decided. Silent defaults become fake physics.

- 1. Is there a privileged reference clock, or only mappings?
- 2. Which layers may an authority change: display, rate, experienced duration, causal graph, settlement?
- 3. Do changes apply prospectively, retroactively in-place, or by branch?
- 4. What can a token grant, and which jurisdiction recognizes it?
- 5. What evidence survives rewind or branch, and who may inspect it?
- 6. Dispute rule: rule-at-event, rule-at-agreement, or rule-at-adjudication?
- 7. Cross-territory effects: immediate, signal-limited, or special mechanism?
- 8. Do bodies and processes follow local rate, reference rate, or a third aging rule?
- 9. Is the past mutable, forkable, or presentation-only editable?
- 10. Are residuals allowed to remain permanent?
- 11. Which of I1–I15 survive every fictional power?

Independent recommendation, still draft: events and original bindings immutable; interpretations, mappings, and settlements revisable and append-only; fictional rewind or branch only as a labeled topology operation; no master timeline until a settlement act needs one. This does not force the fictional universe to have an immutable past. It forces the research system to remember what was asserted, observed, believed, changed, and authorized.

15. Research phases

- Phase 0 — Fill the decision register. Do not implement against silent meanings of time, event, purchase, change, or history.
- Phase 1 — Ontology, glossary, Event Claim Record schema, valid and invalid examples.
- Phase 2 — Deterministic reference graph engine: versioned mappings, append-only claims, branches, causal constraints, point-in-time reconstruction. The reference model is an oracle, not production architecture.
- Phase 3 — Deterministic suites A–E with fixtures and expected-results files. Every failed invariant yields a minimal counterexample.
- Phase 4 — Generated sequences and adversarial injection against the invariants.
- Phase 5 — Multi-contributor challenge on the same schema, corpus, and questions. Compare assumptions and residuals, not prose preference.
- Phase 6 — Canon integration only after testing. Keep experimental results and dissent as provenance.

16. Unresolved paradoxes (keep visible)

- Two frames can disagree on order of spacelike-separated events; demanding one order is demanding a convention.
- An event-driven world has no duration between events; contracts written in hours have no native meaning there.

- A complete audit of every change requires a closed universe of territories and records.
- If presentation can be rewritten faster than inhabitants can sense, “what time they thought it was” becomes another contested archive.
- Purchasing a mapping is not purchasing the processes that already ran under the old mapping.

17. What testing this in full means

Full test is not a longer essay. Full test is: instantiate the schema on a small closed world (two territories, three clocks, one branch point, one fold, one pause, one distant observer); run T1–T10; issue one scoped $\Omega 100$ token and publish S; issue one interval object and attempt a reclock; issue one ly-denominated contract and an origin-definition change; write difference tables; keep residuals. Only then decide which objects are promoted toward canon.

18. Verdict

Use the team packet as a constraint harvest, not as a source to average. Adopt ChatGPT’s stack as a checklist of layers. Adopt Claude’s append-only instinct and extend it with ruleset-time and bindings. Reject Gemini’s “highest permission wins equals truth.” Adopt Perplexity’s insistence on naming asset class, leaving architect decisions open, separating test tracks, and requiring oracles. Reject Copilot’s early master timeline. Keep prior Grok’s four unit kinds, validation lineage, conversion-and-dispute surface, and dependency cone.

The independent statement: time in the lab is a graph of bindings and mappings under jurisdiction, queried by explicit questions, with residuals allowed to survive. That is enough to start the battery. It is not yet canon.

— *End of draft. Not canon. Not a claim on any live system.* —